

The Best Use Cases for AI to Reduce NPT in Drilling and Well Delivery



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Contents

1	Where Is Time Actually Lost?	1
2	Implemented Use Cases	1
2.1	Equipment Health: AI-Driven Component Life Optimization	1
3	Invisible Lost Time (ILT)	3
3.1	A Deployed ILT Use Case: Occidental (IADC/SPE-199616-MS)	4
4	Early Warning for Downhole Events (Stuck Pipe, Losses, Kicks)	5
4.1	Physics-Informed Stuck Pipe Prediction (Liu et al., <i>Petroleum Science</i> , 2026)	5
4.2	SLB: Retrieving Operation Insights with GenAI LLM (ADIPEC 2024)	6
5	Industry Context: Aramco’s METABRAIN LLM	8
5.1	What It Is	8
5.2	What Officials Said Later	8
5.3	Scale and Value Claims	9
6	Conclusion and Recommendation	10
	References	11

List of Figures

1	Mud pump rendering with circles indicating the location of vibration sensors. The red circles indicate potential anomaly.	2
2	Component table showing the predicted lifetimes of mud pump consumables, providing guidance to replace components before they reach their end-of-life. . .	2
3	Rig overview showing equipment conditions. The user can click on a card to see the details of a particular equipment.	3
4	Illustrative graphic: digital monitoring of offshore drilling operations.	3
5	Snapshot of a typical drilling connection divided into three stages: pre-connection (orange), slip-to-slip (dark green), and post-connection (purple).	4
6	Average drilling connection time for each rig. A gap of more than 3 minutes was observed between the best and the worst rig, which presents a significant improvement potential.	4
7	Schematic diagram of the integrated methodology.	5
8	Model structure diagram.	6
9	Precision and recall evaluation (weighted—emphasize on major class; unweighted—treat all classes equally). The red line represents the engineer workflow acceptance threshold, which should be greater than 85%.	7
10	LLM sequential customization workflow (Custom FM is not inclusive in the scope). . .	7

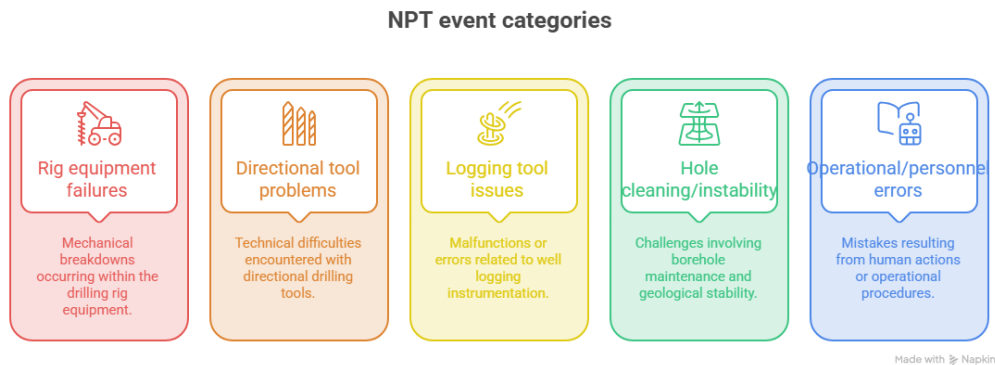
List of Tables

1	Stuck pipe prediction performance on the test wells.	6
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1 Where Is Time Actually Lost?

Operators and contractors put unplanned time at roughly 20–30% of total offshore well time. Shell’s Eric van Oort cites 20–30%, and notes tool failure rising even as lost circulation, instability and well control improved. Maersk Drilling put NPT at about 20% of well construction time, with a large share from downhole problems.

A Utah FORGE study of 550+ NPT events found five categories drove over 60% of NPT: rig equipment failures, directional tool problems, logging tool issues, hole cleaning/instability, and operational/personnel errors [13]. That is geothermal, but the pattern matches oil and gas.



Five NPT event categories that together drove over 60% of NPT in the Utah FORGE study.

Beyond recorded NPT, there is **invisible lost time (ILT)**. ILT is the gap between actual and benchmark time for an operation, e.g. slow connections. It never shows up in the NPT column, so most companies undercount it.

2 Implemented Use Cases

2.1 Equipment Health: AI-Driven Component Life Optimization

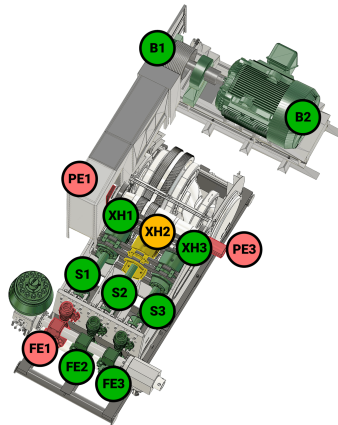
Precision reduces equipment cost of ownership through AI-driven component life optimization [14].

Targeting high-impact areas. Implementation began with mud pumps rather than top drives due to the high frequency of fluid end consumable failures and substantial yearly replacement costs.

Using AI-driven analytics, Precision has been able to achieve a 54% extension of mud pump fluid end component life.

Overcoming variable load challenges. Traditional sensor thresholding fails in drilling because dynamic load changes (e.g. standpipe pressure jumps) alter vibration readings. The solution incorporates over 30 real-time operational parameters (pressure, speed, mud flow, volume) alongside vibration metrics, using genetic optimization algorithms to handle the massive parameter space.

Remaining Useful Life (RUL) modeling. Using a custom Transformer deep learning architecture trained on 96 billion data points, the system predicts individual component end-of-life



PUMP HOURS PER COMPONENT					
STATUS	COMPONENT	LIFETIME	[%]	PUMP HOURS	LIFETIME HOURS
⚠️	Disch. Valve 3		93%	2,518	2,700
⚠️	Suction Seat 1		89%	2,518	2,800
⚠️	Suction Seat 2		89%	2,518	2,800
⚠️	Suction Seat 3		89%	2,518	2,800
⚠️	Suction Valve 1		89%	2,518	2,800
⚠️	Suction Valve 2		89%	2,518	2,800
⚠️	Suction Valve 3		89%	2,518	2,800
⚠️	Disch. Seat 2		88%	2,518	2,850
✅	Disch. Seat 1		73%	922	1,250
✅	Disch. Valve 1		73%	922	1,250
✅	Disch. Seat 3		71%	1,747	2,450
✅	Disch. Valve 2		67%	1,219	1,800

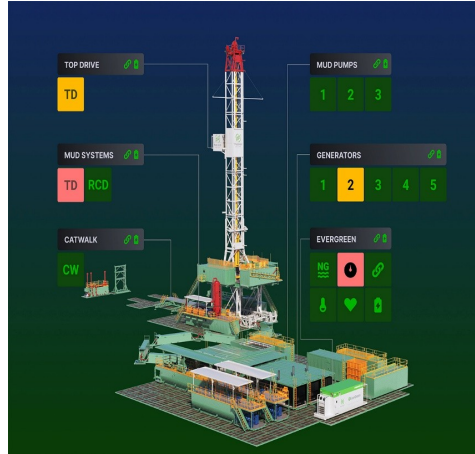


Figure 3. Rig overview showing equipment conditions. The user can click on a card to see the details of a particular equipment.

weeks or months in advance with over 90% accuracy (within $\pm 5\%$ of actual lifetime).

Vendor quality evaluation. The AI normalizes variable drilling conditions across different basins, enabling objective performance comparisons of component vendors.

Demonstrated Results

- **Failure & NPT reduction:** Predicts up to 88% of fluid end failures, yields an 85% reduction in module failures, and reduces mud pump Code-8 Non-Productive Time (NPT) by 75%.
- **Efficiency & cost savings:** Extends mud pump fluid end component life by 54%, achieves a 38% reduction in overall maintenance costs, and yields a 28% improvement in Rate of Penetration (ROP).
- **Fleetwide scale:** The asset health digital twin processes 3 billion data points daily across 120+ rigs in the US, Canada, and the Middle East, monitoring over 7,000 components.

3 Invisible Lost Time (ILT)



Figure 4. Illustrative graphic: digital monitoring of offshore drilling operations.

The problem. Unplanned time is often cited at about 20–30% of offshore well time (Shell’s Eric van Oort puts it at 20–30%). On top of recorded NPT sits **invisible lost time (ILT)**: slow connections, trips and suboptimal parameters that never reach the NPT column. AI helps most where data is high-frequency, failures are repeatable and crews can act on the output.

3.1 A Deployed ILT Use Case: Occidental (IADC/SPE-199616-MS)

A rule-based algorithm detects every drilling and tripping connection from one-second surface data, then splits each into pre-connection, slip-to-slip and post-connection [1].

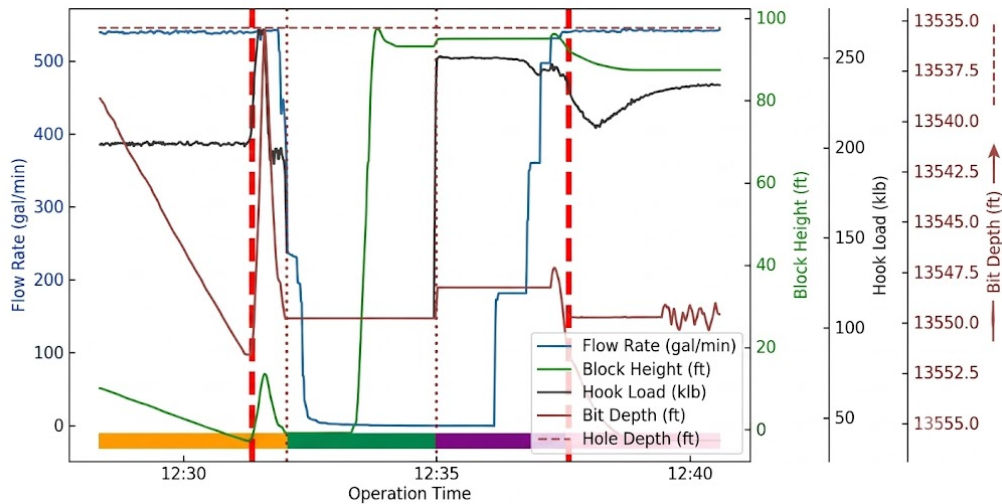


Figure 5. Snapshot of a typical drilling connection divided into three stages: pre-connection (orange), slip-to-slip (dark green), and post-connection (purple).

Run on roughly 370 historical wells, it found a gap of more than 3 minutes between rigs of the same contractor on similar wells (about 6 min for best performers vs. an 8 min average).

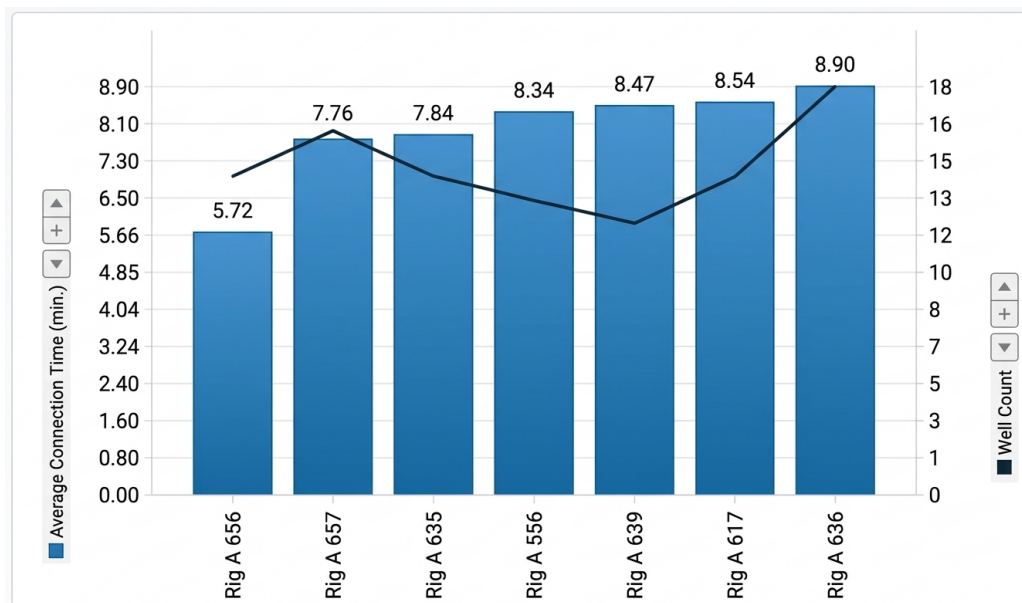


Figure 6. Average drilling connection time for each rig. A gap of more than 3 minutes was observed between the best and the worst rig, which presents a significant improvement potential.

The tool was deployed on an IoT edge device at the rig, with a simple driller screen showing

the last connection, crew average and rig comparison. On 140 wells after deployment, drilling connection time fell about 1.5 min in one business unit and over 2 min in another. The company reports about 6 hours saved per well and about \$2.4M potential annual savings in one unit.

The authors credit much of the gain to the real-time driller display, not the algorithm alone. Visibility drove behavior change.

4 Early Warning for Downhole Events (Stuck Pipe, Losses, Kicks)

4.1 Physics-Informed Stuck Pipe Prediction (Liu et al., *Petroleum Science*, 2026)

Problem. Stuck pipe is a major NPT driver. Pure physics models need inputs that are often unavailable in real time, and pure ML models trained on scarce incident labels overfit and give physically implausible results, with false alarms and missed detections in the field [2].

Approach. A hybrid model with two parts:

- A drag-torque model and a hydraulic model compute theoretical hook load, torque and standpipe pressure. **Bayesian optimization** recalibrates the uncertain parameters from the live data stream every 30 minutes: friction factor and pipe weight factor for drag-torque, and the rheology parameters (n , K , τ_y) for hydraulics. The search space is bounded to physically plausible values, e.g. friction factor between 0.1 and 0.6.
- An unsupervised **LSTM autoencoder** monitors eight physics-derived features covering three sticking types: mechanical, differential and hole-cleaning/pack-off. It is trained only on normal drilling data, so it needs no labeled stuck-pipe events. A rising reconstruction error signals risk.

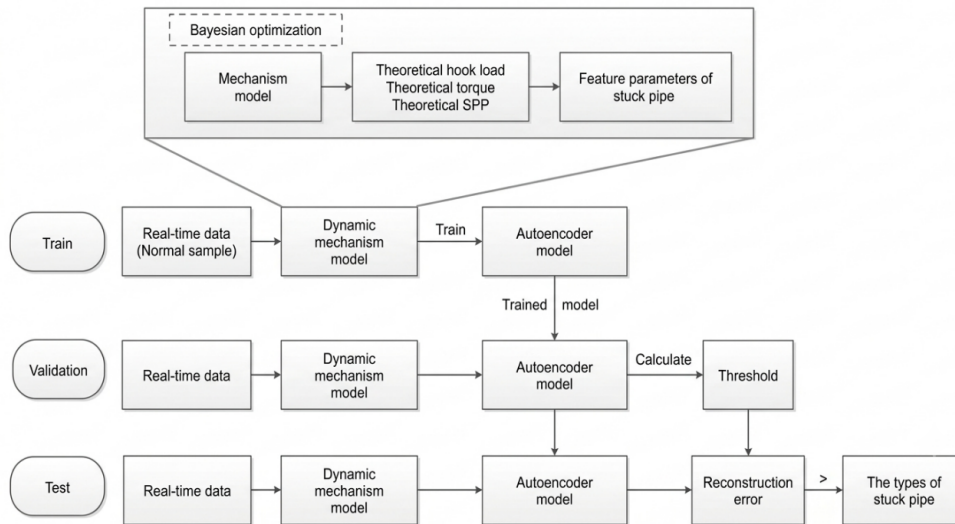


Figure 7. Schematic diagram of the integrated methodology.

Alarm design. The threshold is set so that missed detections are minimized, because a missed stuck pipe costs far more than a false alarm. The threshold was 0.93 on the validation well.

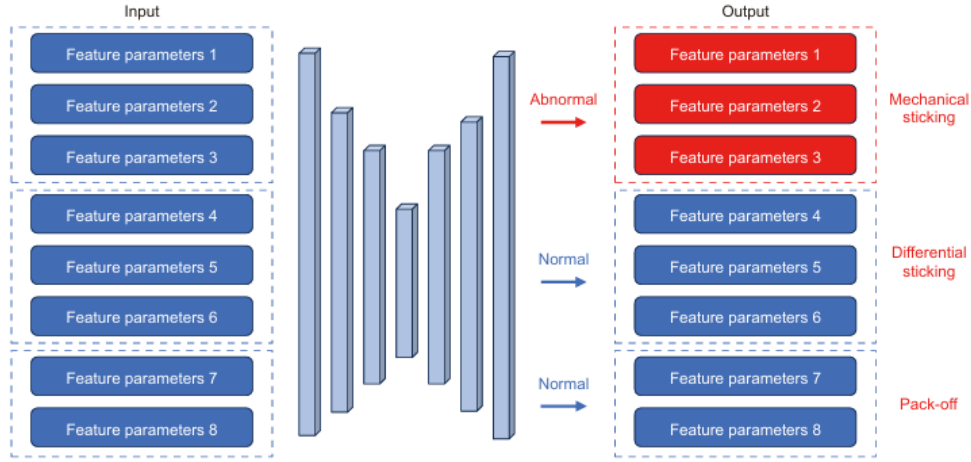


Figure 8. Model structure diagram.

Data. 46 wells (37 onshore, 9 offshore) in Chinese oilfields with 51 stuck pipe incidents, 45 of them during tripping. 14 wells were used for testing.

Table 1. Stuck pipe prediction performance on the test wells.

Model	Accuracy	False alarm	Missed detection	Avg. warning
Drag-torque (physics only)	68.5%	31.2%	31.5%	8.5 min
Empirical rules	72.3%	27.5%	27.7%	10.1 min
LSTM (supervised)	83.7%	16.1%	16.3%	16.8 min
LSTM autoencoder (raw data)	85.4%	14.5%	14.6%	18.7 min
Hybrid (this paper)	92.5%	7.3%	7.5%	22.5 min

Results versus baselines (test wells).

4.2 SLB: Retrieving Operation Insights with GenAI LLM (ADIPEC 2024)

Problem. Risks that cause NPT are often buried in free-text remarks in daily drilling reports (DDRs). Report formats vary by operator and vendor, and manual extraction is slow [3].

Data. 178 DDR files from Equinor’s public Volve dataset. A drilling engineer validated 13 risk classes plus “no risk” over about 1,740 labelled rows, with 342 held out for testing.

Method. The authors built the workflow in stages and compared each:

- Zero-shot GPT-4 scored 61% accuracy.
- One-shot prompting with an explanation for each label reached 81%.
- GPT-4 with retrieval (RAG) over 22,000 DDR remarks reached 85%.
- Fine-tuning small open models with QLoRA, plus in-house domain “safeguard” post-processing, was the best route.

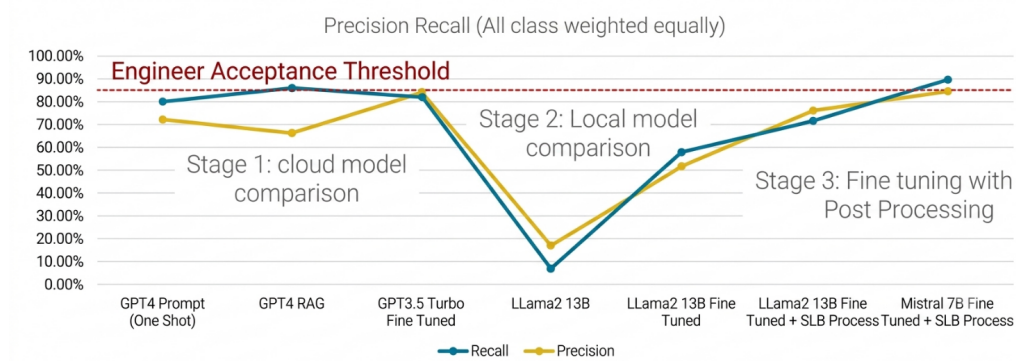


Figure 9. Precision and recall evaluation (weighted—emphasize on major class; unweighted—treat all classes equally). The red line represents the engineer workflow acceptance threshold, which should be greater than 85%.

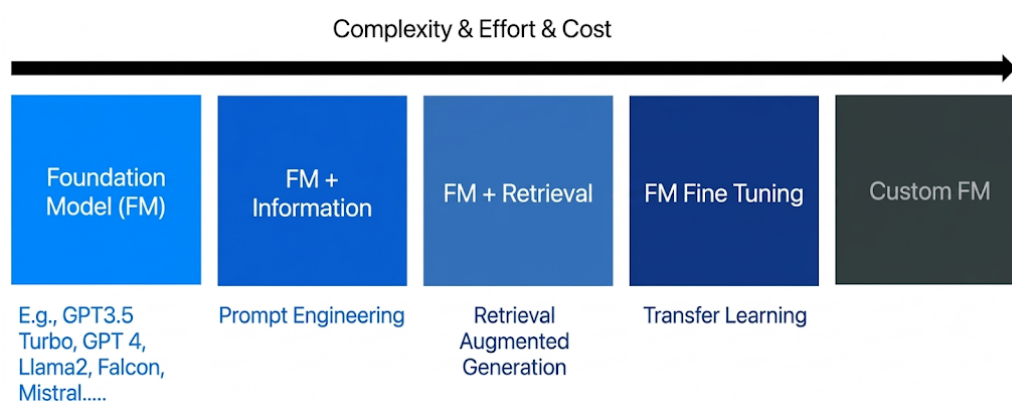


Figure 10. LLM sequential customization workflow (Custom FM is not inclusive in the scope).

Best result. Fine-tuned Mistral 7B with post-processing reached 94% accuracy, with unweighted precision of about 85% and recall of about 90%. It beat fine-tuned GPT-3.5 Turbo (90%) and the earlier fine-tuned GPT-2 benchmark. The GPT-2 model scored 89% accuracy but only about 47% unweighted precision, so it missed rare risk classes.

Engineering threshold. Drilling engineers set 85% as the minimum acceptable reliability for automated risk identification.

Practical design choices.

- The 7B model ran on a single on-premises A100 GPU, which addresses data-residency concerns.
- Smaller models matched larger ones once they were fine-tuned and constrained with domain rules.
- Chunk size and glossary choice affected RAG quality. Real DDR remarks worked far better than generic glossaries as a knowledge base.

Field application. The model ran on 180 DDRs from 24 wells. It surfaced roughly 7% hidden events in remarks not captured as NPT events (the paper's wording here is ambiguous, so the figure is approximate). It found about 2% more events than manual reading. Review time dropped from about a month (about 6 reports a day) to roughly 1.5–2 hours.

Downstream use. Outputs feed geomechanics (post-drill 1D mechanical earth model calibration), drilling-engineer planning, and formation-level risk maps for infill well placement.

Why it matters for NPT. This is a pre-drill and post-drill learning tool. It finds recurring and latent risks before the next well, rather than cutting NPT directly.

5 Industry Context: Aramco's METABRAIN LLM

5.1 What It Is

At LEAP in March 2024, Aramco unveiled Metabrain, a generative AI model it described as a pioneering industrial technology, trained on about 90 years of company data. CEO Amin Nasser said it was helping upstream teams analyse drilling plans, geological data, and historical drilling time versus cost, and recommend well options. Nasser said Metabrain has the potential to raise productivity and growth and transform the way Aramco works.

5.2 What Officials Said Later

At the GAIN summit in September 2024, EVP Technology & Innovation Ahmad Al-Khowaiter said Aramco's AI models were already analysing drilling plans and geological data to recommend the best well options for engineers. In November 2025, he described it as the energy industry's first industrial LLM, "which we use in our day-to-day business." At Davos in January 2026, Nasser said, "Our goal is to make the energy industry smarter through AI," and stressed that data quality is critical. He also said Aramco's advantage lies in nearly a century of geological and operational data to train and refine models.

5.3 Scale and Value Claims

Aramco reported \$1.8 billion of AI-driven “technology realized value” in 2024, and Nasser expected \$3–5 billion for 2025. It has identified 442 AI use cases, with more than 200 deployed. Aramco defines this value as gains from lower capital spending, higher revenues or reduced operating costs from AI and digital tools.

6 Conclusion and Recommendation

AI reduces NPT best when it is matched to where time is actually lost, and the evidence differs sharply by use case.

Proven today

Equipment health monitoring and invisible-lost-time analytics. Occidental's real-time connection-time tool cut drilling connections by about 1.5 minutes, which the company estimates at roughly 6 hours saved per well. These use cases need modest data, deliver fast payback, and gain from crew visibility.

Promising, with care

Early warning for stuck pipe and other downhole events. A 2026 hybrid physics-plus-ML model reported 92.5% accuracy, a 7.3% false-alarm rate and a 22.5-minute average warning on 14 test wells. Results like these still need prospective field proof, and alarm management will decide whether crews trust them.

Emerging

GenAI for learning from offset wells. SLB's fine-tuned 7B model classified drilling-report risks at 94% accuracy, on one public field. It supports planning and post-well learning more than real-time NPT cuts.

Next horizon

Closed-loop automation and multimodal supervisors, which cut NPT mainly through consistent execution.

Recommended Path

1. **Start with data and ILT** (rig-state labels, consistent NPT coding, connection and tripping KPIs).
2. **Add equipment health and advisory downhole alerts**, tuned for precision and warning lead time.
3. **Layer in GenAI knowledge tools and automation** once trust and data quality are proven.

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